

PABLO UBILLA PAVEZ

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EDUCATION

PhD	Machine Learning — INRIA & Université de Montpellier
2025–Now	Thesis on <i>Deep Learning Methods for Species Distribution Models</i> . Supervised by Diego Marcos & Christophe Botella.
MSc	Ecology and Data Science — University College London
2023–2024	Degree awarded with Highest Honours (Top 5%). Thesis on <i>Consensus Clustering for Functional Groups</i> . Supervised by Daniel Maynard.
Master	Operations Management — Universidad de Chile
2022–2023	Degree awarded with Highest Honours (Ranked 1/15). Thesis on <i>Outlier Detection for Elections with Demographic Groups</i> . Supervised by Charles Thraves.
Prof. Degree & BSc	Industrial Engineering — Universidad de Chile
2017–2023	Degree awarded with Highest Honours (Ranked 1/174). Exchange period at Universiteit Leiden (Spring 2022).

RESEARCH INTERESTS

Machine Learning, Statistics, and Ecology.

PUBLICATIONS

2026	Ubilla Pavez, P., Paz, A., & Maynard, D. S. <i>Functional group classification using consensus clustering</i> . PLOS Computational Biology, 22(5), 1–25. doi:10.1371/journal.pcbi.1014278
2024	Thraves, C., Ubilla Pavez, P., & Hermosilla, D. <i>A Fast Ecological Inference Algorithm for the $R \times C$ case</i> . Under revision. Preprint available at SSRN 4832834.

SOFTWARE

[faitei](#): Ecological inference methods (R package in CRAN, 2025). Work with Thraves, C., and Hermosilla, D.

WORK EXPERIENCE

Oct 2024 – Sep 2025	Airnguru , <i>Data Scientist</i>
Remote, Chile	Developed automation of large-scale pricing strategies (up to billions of fares).
Jan 2021 – Jan 2022	Complex Engineering Systems Institute , <i>Research Assistant</i>
Santiago, Chile	Adaptive question selection using dynamic programming, greedy algorithms, and reinforcement learning to maximize the information obtained at the end of a survey.
Sep 2022 – Jan 2023	Municipalidad de Vitacura , <i>Data Scientist</i>
Santiago, Chile	Estimated daily street-level recycling collection and optimized truck routes using GPS historical data, leading to a redesigned partition of municipal collection sectors.
Aug – Oct 2020	Fundación Tierra Viva , <i>Intern</i>
Santiago, Chile	Contributed in data collection for <i>Resumen del Estado del Medio Ambiente 2020</i> .

TEACHING EXPERIENCE

*Seasons correspond to the hemisphere of the institution. †TA: Teaching Assistant; EA: Exam Assistant; I: Instructor.

FRB-CESAB

June 2026 **Introduction to Deep SDMs (I)**, AI for Ecologists: a toolkit

University College London

Fall 2023	Financial Engineering (TA) , MSc Computational Finance
Spring 2024	Applied Computational Finance (TA) , MSc Computational Finance

Universidad de Chile

- Fall 2023 **Modelling and Optimization** (TA), Industrial Engineering
- Fall 2020 **Modelling and Optimization** (TA), Industrial Engineering
- Spring 2022 **Decisions Under Uncertainty** (TA), Industrial Engineering
- Spring 2021 **Decisions Under Uncertainty** (TA), Industrial Engineering
- Spring 2021 **Game Theory and Strategy** (TA), Industrial Engineering
- Fall 2021 **Probabilities** (TA), Industrial Engineering
- Fall 2021 **Operations Research** (TA), Industrial Engineering
- Spring 2020 **Operations Research** (TA), Industrial Engineering
- Fall 2022 **Probabilities** (EA), Industrial Engineering
- Fall 2022 **Information Engineering** (EA), Industrial Engineering
- Spring 2019 **Modelling and Optimization** (EA), Industrial Engineering
- Fall 2020 **Introduction to Calculus** (EA), Mathematics Engineering
- Spring 2019 **Introduction to Algebra** (EA), Mathematics Engineering
- Fall 2019 **Probabilities and Statistics** (EA), Mathematics Engineering

Universidad Finis Terrae

- Fall 2023 **Stochastic Models** (I), Industrial and Informatics Engineering

Universidad Adolfo Ibáñez

- Summer 2022 **Operations Research** (TA), Industrial Engineering
- Spring 2021 **Operations Research** (TA), Industrial Engineering

AWARDS & SCHOLARSHIPS

- 2024 **Best Presentation Prize**, MSc Ecology and Data Science
- 2024 **Dean's List**, MSc Ecology and Data Science
- 2024 **Distinguished Master's Graduate Award**, Postgraduate Engineering School UCH
- 2023 **Masters Abroad Scholarship**, *fully funded*, ANID
- 2022 **Academic Excellence Scholarship**, MSc Operations Management
- 2022 **International Mobility Scholarship**, Santander Bank
- 2018–2021 **Outstanding Student**, Faculty of Physics and Mathematics UCH

TALKS & POSTERS

Deep Learning Integration for Species Distribution Models

- 2026 **World Biodiversity Forum** (*Davos, Switzerland*) (Talk)
- 2026 **The European Conference for Biodiversity Monitoring** (*Montpellier, France*) (Talk)
- 2026 **Rencontre CiSStats-RESSTE** (*Avignon, France*) (Talk)

Consensus Clustering for Functional Groups

- 2026 **INRIA PhD Seminar** (*Montpellier, France*) (Talk)
- 2024 **British Ecological Society: Macroecology** (*Cardiff, UK*) (Talk)

A Fast Ecological Inference Algorithm for the RxC case

- 2025 **Workshop in Management Science** (*Pucón, Chile*) (Poster)
- 2024 **Operational Research Society 66** (*Bangor, UK*) (Talk)

Optimized sensor deployment in biodiversity monitoring

- 2024 **Surrey Wildlife Trust Showcase** (*Surrey, UK*) (Talk)

DEVELOPMENT

- Summer 2023 **XVIII Summer School in Discrete Mathematics** — ISCV
Valparaíso, Chile High-chromatic graphs; provable algorithms for machine learning; linear programming.
- Winter 2020 **9th Winter School in Engineering Systems** — Universidad de Chile
Santiago, Chile Stackelberg games; AI challenges in big data; urban mobility.

SKILLS

Programming	Python, R, SQL, MATLAB	Machine Learning	TensorFlow, PyTorch, Scikit-Learn
Optimization	Gurobi, OR-Tools, CVXPY	Other	Git, L ^A T _E X, HPC